ABSTRACT

With the upcoming Paris 2024 Olympics, social media platforms are expected to experience a surge in global interactions, which can lead to an increase in hate speech due to varying cultural, political, and personal viewpoints. Detecting hate speech in this setting involves analysing both text and emojis, as emojis can modify or intensify the tone of messages. Addressing this challenge is essential to maintain a positive online environment and to ensure respectful exchanges during the international event.

This project aims to develop a context-aware tweet classification system using Long Short-Term Memory (LSTM) and Bidirectional LSTM (Bi-LSTM) networks. The project focuses on preprocessing the data, which includes data cleaning, emoji-text association, lemmatization, handling URLs and mentions, tokenization, and the removal of stop words. The dataset is then labelled into Hate and Non-Hate categories, with special attention to the role of emojis in prediction. Feature extraction techniques like word embedding (GloVe), emoji embedding, and N-grams (bi-grams) are used to represent the tweet's contextual features. The system is implemented using LSTM and Bi-LSTM architectures, both with and without clustering and Attention Mechanism. The evaluation metrics for the project include accuracy, precision, recall, F1-score, and others. The highest accuracy achieved with this model is 98%, showcasing the effectiveness of the context-aware approach.

The system is developed on the Jupyter Notebook platform, which provides an interactive environment for data exploration, model training, and testing. Jupyter Notebook’s integration with Python and compatibility with popular machine learning libraries such as TensorFlow and PyTorch make it suitable for implementing and fine-tuning complex models like LSTM and Bi-LSTM.

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